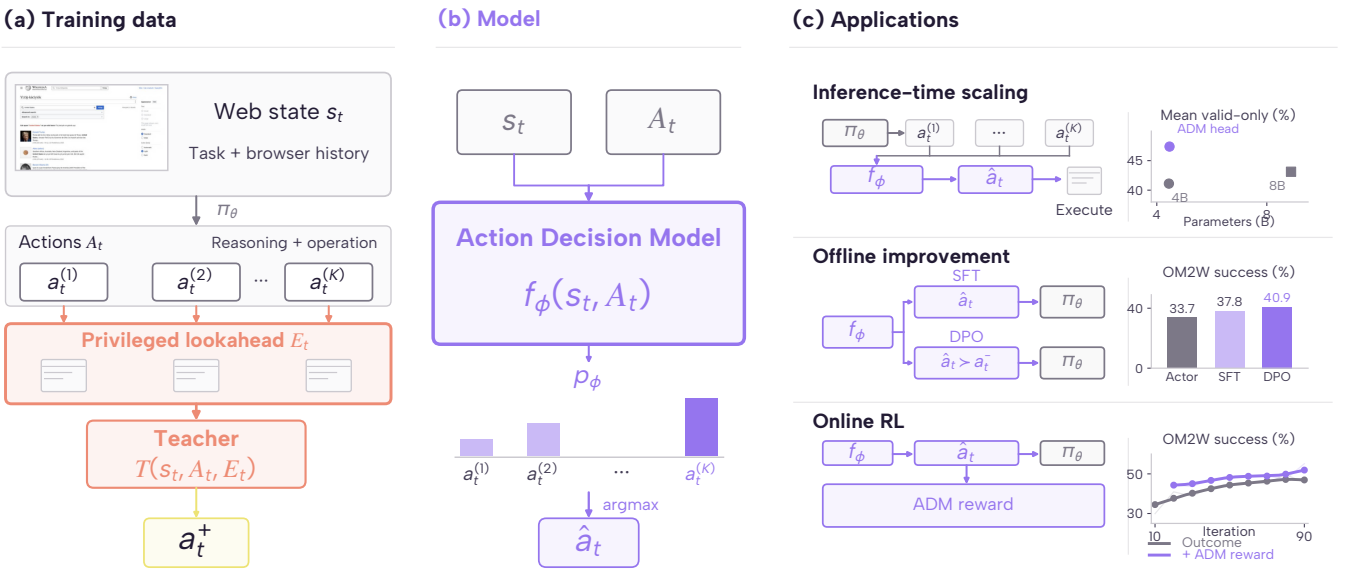
(a) Training data
Web state s_t
Task + browser history
π_θ
Actions A_t
Reasoning + operation
$a_t^{(1)}$
$a_t^{(2)}$
...
$a_t^{(K)}$
Privileged lookahead E_t
Teacher
$T(s_t, A_t, E_t)$
a_t^+
(b) Model
s_t
A_t
Action Decision Model
$f_\phi(s_t, A_t)$
p_ϕ
$a_t^{(1)}$
$a_t^{(2)}$
...
$a_t^{(K)}$
argmax
$\hat{a}_t$
(c) Applications
Inference-time scaling
π_θ
$a_t^{(1)}$
...
$a_t^{(K)}$
f_ϕ
$\hat{a}_t$
Execute
Mean valid-only (%)
ADM head
45
40
4B
8B
4
8
Parameters (B)
Offline improvement
SFT
f_ϕ
$\hat{a}_t$
π_θ
DPO
$\hat{a}_t > a_t^-$
π_θ
OM2W success (%)
40
0
33.7
37.8
40.9
Actor
SFT
DPO
Online RL
f_ϕ
$\hat{a}_t$
π_θ
ADM reward
OM2W success (%)
50
30
10
Iteration
90
Outcome
+ ADM reward